

Examination Scheme for EoSE for Semester-II

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Duration of Examination		Maximum Marks		Minimum Marks	
Theory	BTH-52T-251 MOLECULAR BIOLOGY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-52P-252 MOLECULAR BIOLOGY-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH- 52T-253 BIOINFORMATICS AND BIOSTATISTICS	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-52P-254 BIOINFORMATICS AND BIOSTATISTICS-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH- 52T-255 INSTRUMENTATION AND BIOTECHNIQUES	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-52P-256 INSTRUMENTATION AND BIOTECHNIQUES-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks

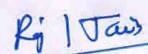
The theory question paper will consist of **two** parts A&B.

PART-A: 20 Marks

Part A will be compulsory having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

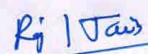
PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with internal choice. Each question will carry 15 marks.


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UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)								
				Bio-Technology 1 st YEAR SEM-II			Credits	
#	Level	Semester	Type	Title	L	T	P	Total
1.	5	II	MJR	UG0804 -BTH-52T-251 MOLECULAR BIOLOGY	4	0	0	4
2.	5	II	MJR	UG0804 BTH-52P-252 MOLECULAR BIOLOGY - PRACTICAL	0	0	2	2
3.	5	II	MJR	UG0804 BTH- 52T-253 BIOINFORMATICS AND BIOSTATISTICS	4	0	0	4
4.	5	II	MJR	UG0804 -BTH-52P-254 BIOINFORMATICS AND BIOSTATISTICS -PRACTICAL	0	0	2	2
5.	5	II	MJR	UG0804 -BTH- 52T-255 INSTRUMENTATION AND BIOTECHNIQUES	4	0	0	4
6.	5	II	MJR	UG0804 -BTH-52P-256 INSTRUMENTATION AND BIOTECHNIQUES -PRACTICAL	0	0	2	2

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
II	BTH-52T-251	MOLECULAR BIOLOGY			5	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Introductory	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Biology Courses of Senior Secondary level				


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Objectives of the Course	➤ To learn and understand the important discoveries that are made in the field of molecular biology. ➤ To learn key molecular events that occur during the DNA replication, transcription, translation and regulation of gene concept. ➤ gain knowledge on the foundation of genetic engineering and their applications in biological research as well as in biotechnology industries.
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Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ Learn and understand the important discoveries that are made in the field of molecular biology. ➤ Learn key molecular events that occur during the DNA replication, transcription, translation and regulation of gene concept.
2. Memorizing	➤ Understand molecules involved in cell functioning and their importance.
3. Applying	➤ Acquainted with gene organization and regulation of gene expression and its importance in biology

Detailed Syllabus

BTH- 52T-251 MOLECULAR BIOLOGY

Unit-I

Genes and DNA: Genome, Gene Double helical structure of DNA, DNA supercoiling, Gene structure, Non-coding DNA and RNA. **DNA Replication:** Mechanisms of prokaryotic DNA replication: Initiation, Elongation and Termination; DNA Polymerases, Helicase, other enzymes and accessory proteins involved in DNA replication. Fidelity of replication and coordinating synthesis of the leading and lagging strands, Okazaki fragments.

15 Lectures

Unit-II

DNA Damage and Repair: Causes of DNA damage and molecular mechanisms of repair, excision repair system in bacteria and eukaryotes, base excision, recombination repair systems and SOS repair. Transcription and post transcriptional changes.

15 Lectures

Unit-III

Transcription: Types of RNA, mRNA structure, prokaryotic and eukaryotic RNA polymerases, transcriptional factors, Promoter sequences, binding sites for RNA polymerases, transcription initiation, elongation, termination, attenuation and antitermination.

15 Lectures

Unit-IV



Translation: Structure of tRNA, ribosome, genetic code, translation formation of initiation complex, initiation factors and their regulation elongation and elongation factors, aminoacylation of tRNA, aminoacyl tRNA synthetases, and termination in Prokaryotes and eukaryotes. Co and post-translational modifications of proteins.

15 Lectures

BTH- 52P-252 MOLECULAR BIOLOGY -PRACTICAL

1. Building of a model of B-DNA.
2. Isolation of Plant DNA and its quantification by spectrophotometric method.
3. Separation and visualization of DNA fragments by Agarose gel electrophoresis.
4. Demonstration of SDS-PAGE.
5. Determination of denaturation and renaturation of DNA double helix.
6. Isolation of RNA and quantification by spectrophotometric method.
7. Polymerase chain reaction.
8. Southern blot analysis using a gene specific probe.

Suggested readings:

1. Lewis, B. (2001). **Genes X**, Oxford University Press, New York.
2. Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K. and Watson, J.D. (1999). **Molecular Biology of the cell**. Garland Publishing Inc. New York.
3. Wolfe, S.L. (1993). **Molecular and Cellular Biology**. Wadsworth Publishing Co., California, USA.
4. Lodish, H., Berk, A., Zipursky, S.L., Matsudaira, P., Baltimore, D., And Damell, J. 2000. **Molecular Cell Biology** (4th Edition). W.H. Freeman and Co., New York, USA.
5. Glick, B. R. and Thompson, J.F. (1993). **Methods in Plant Molecular Biology and Biotechnology**, CRC press Boca Raton, Florida.
6. Malacinski, G.M. and Friedfirlder, D. (1998). **Essentials of Molecular Biology** (3rd Edition). Jones and B Artlet Publishers, inc. London.
7. Jordanka Zlatanov. (2023). **Molecular Biology: Structure and Dynamics of Genomes and Proteomes** (2nd Edition).

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Develop sufficient knowledge about the characteristics of the genetic material and structure of DNA and RNA.
2. Recognize DNA organization in chromosomes and molecular mechanism of DNA replication, and transcription.
3. Understand characteristic and importance of genetic code and molecular mechanism of translation.
4. Understand molecular structure of the gene and regulatory mechanisms for gene expression.
5. Understand the structure, function, and variations in DNA and RNA.
6. Have insights into the various models for chromatin organization.
7. Understand step wise processes of replication, transcription and translation.
8. Develop knowledge and understanding of the operon concept and gene regulation mechanisms.
9. Understand the role of protein and its modification in DNA packaging.
10. Have knowledge about the differentiation of molecular mechanism of replication, transcription and translation.
11. Understand the role of protein/transcriptional factor in gene regulation.
12. Develop acumen about the variation in gene regulation processes.
13. Use scientific methods, and critical thinking skills to ask questions and solve problems.


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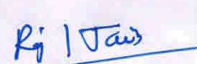
BTH-52T-253 BIOINFORMATICS AND BIOSTATISTICS

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
II	BTH-52T-253	BIOINFORMATICS AND BIOSTATISTICS			5	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Introductory	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Biology Courses of Senior Secondary level				
Objectives of the Course		➤ The aim of this module to provide practical training in bioinformatics including accessing the major public sequence databases, use of the different computational tools to find sequences, analysis of protein and nucleic acid sequences by various software packages				

Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ Various resources or tools available for bioinformatics. ➤ Sequence similarity and alignment using the bioinformatics tools.


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	➤ The principal concepts about biostatistics. ➤ Compute statistical problems using computer and graphical means. ➤ Solve mean and variance of discrete and continuous distribution.
2. Memorizing	➤ Sequence alignment for various molecules and phylogenetic analysis.
3. Applying	➤ The students will be knowing the diagnostic procedures and collection and processing of specimen. ➤ Appraise statistical test, t distribution and the standard error formulas.

Detailed Syllabus

BTH-52T-253 BIOINFORMATICS AND BIOSTATISTICS

UNIT-I

Concept of Bioinformatics: Introduction and future prospects; Applications in genomics and proteomics; Public databases; Gene bank: Database searches; sequence retrieval systems; similarity searching; BLAST, FASTA; Multiple sequence alignment: database and online tools: Biological databases: Types and applications; Sequence databases: GenBank, EMBL, DDBJ, PIR-PSD, SWISS PORT; Structure Databases: PDB, SCOP, NDB; Derived Databases:- PROSITE, PRINTS, TIGR **15**

Lectures

UNIT-II

Applications of Bioinformatics, Computational methods for sequence analysis: Dot Blot and dynamic programming methods; Phylogenetic analysis; Virtual and electronic cell, Internet tools for DNA sequence translation; Restriction enzyme mapping; Prediction of secondary structure of proteins; Applications tools- Primer designing, molecular mapping and concept and tools of computer aided drug designing

15 Lectures

UNIT-III

Fundamentals of statistics: Scope of statistics for biological research, types of Data, Arithmetic mean, median, mode: theory and simple numerical problems; Measures of variation: Standard deviation, Variance, coefficient of variation; Correlation, types and methods: simple, multiple, linear and nonlinear correlation, Regression: Linear and curvilinear regression (for two variable X and Y only).

15 Lectures

UNIT-IV

Tests of significance: Null hypothesis; Standard Error, Level of Significance; Degrees of freedom; Significance of mean for large samples; significance of means for small samples (Student t-test); significance in ratio of two samples; F test (for difference between variance of two samples); Chi square test; Analysis of variance test (ANOVA) for one and two way classification. Laws of probability, theorem of total probability **15 Lectures**



BTH-52P-254 BIOINFORMATICS AND BIOSTATISTICS -PRACTICAL

1. Introduction to bioinformatics databases (any three): NCBI/PDB/DDBJ, uniprot, PDB etc.
2. Sequence retrieval using BLAST.
3. Sequence alignment.
4. Protein structure prediction.
5. Prediction of different features of a functional gene.
6. Determination of statistical averages/central tendencies.
 - a) Arithmetic mean b) Median c) Mode
7. Determination of measures of Dispersion
 - a) Mean Deviation b) Standard deviation and coefficient of variation c) Quartile deviation
8. Tests of significance-Application of following
 - a) Chi square test b) T-test c) Standard error
9. To learn graphical representation of statistical data with the help of computers (e.g. MS Excel)

Suggested readings:

1. **Introduction to Bioinformatics**, Arthur M. Lesk, Oxford University press.
2. **Introduction to Bioinformatics**, Attawood, Pearson Education.
3. **A textbook of systems biology**, E. Klipp, W. Leibermester, C. Wierling, Axel Kowals, H., Lehrach, R. Herwig (2009), Wiley VCH GmbH.
4. **Bioinformatics: Sequence and Genome analysis**, David. W. Mount (2001), Cold Spring House Laboratory press.
5. **Plant System biology**, Coruzzi, G.M. (2009) Wiley Publishing House.
6. **Bioinformatics- A Practical guide to the analysis of genes and Proteins**, 2nd Edition by Baxevanis.
7. **Practical Statistics for Experimental Biologist**, Wardlaw, A.C. (1985).
8. **Statistical methods in Biology**, Bailey (2000), N.T.J English Univ Press.
9. **Biostatistics**, 7th Edition by Daniel and Fundamental of biostatistics by Khan.
10. **Introduction to biostatistics**, Le and chap (2009), Wiley and sons.
11. **New frontiers of Biostatistics and bioinformatics** (ICSA Book Series in statistics), 1st Edition 2018, springer.

Course Learning Outcomes:

At the completion of the course, the student would be able to:

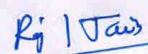
1. Gain an understanding of the concepts and future prospects of bioinformatics, along with its applications in genomics and proteomics.
2. Learn to navigate and utilize public biological databases, including sequence and structure databases like GenBank, EMBL, DDBJ, PIR-PSD, SWISS PORT, PDB, SCOP, NDB, PROSITE, PRINTS, and TIGR.
3. Develop skills in computational methods for sequence analysis, such as Dot Blot and dynamic programming methods, and understand phylogenetic analysis and virtual electronic cell concepts.


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4. Utilize internet tools for DNA sequence translation, restriction enzyme mapping, and protein secondary structure prediction, and gain proficiency in application tools for primer designing, molecular mapping, and computer-aided drug design.
5. Understand and apply fundamental statistical concepts to biological research, including measures of central tendency (mean, median, mode), variation (standard deviation, variance, coefficient of variation), and correlation and regression analysis.
6. Master various tests of significance, including t-tests, F-tests, Chi-square tests, and ANOVA, and apply the laws and theorems of probability to biological data analysis.

BTH-52T-255 INSTRUMENTATION AND BIOTECHNIQUES

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
II	BTH- 52T-255	INSTRUMENTATION AND BIOTECHNIQUES			5	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Introductory	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Biology Courses of Senior Secondary level				


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Objectives of the Course	➤ The objective of the course is to introduce various techniques to the students, which are used in biological research as well as to provide them with an understanding of the underlying principles of these techniques and experimental skills in the form of practical exercises so that students can apply this knowledge to improve their understanding of the subject and better execution of these techniques.
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Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ Various spectroscopic techniques and their application ➤ Define various principles and applications of various chromatography, electrophoresis and of centrifuge techniques. ➤ The principle of various microscopy
2. Memorizing	➤ Define various principles of various techniques
3. Applying	➤ Define various applications of various techniques

Detailed Syllabus

BTH-52T-255 INSTRUMENTATION AND BIOTECHNIQUES

UNIT-I

Buffers- Preparation and principle of pH meter

Microscopy – Principle and application of Dissecting and compound Microscope, phase contrast, Fluorescence and Electron microscopy (SEM and TEM)

Spectroscopy: basic principle, instrumentation, application, UV visible spectrophotometer

IR & Raman spectroscopy – Basic principle, theory and qualitative interpretation of I.R. spectra, quantitative methods

15 Lectures

UNIT-II

Fluorescence spectroscopy- Principle, Instrument Design, Methods & Applications

Centrifugation & Ultracentrifugation- Basic principles, Forces involved, RCF Centrifugation, techniques-principal, types and applications.

Chromatography- Basic Concepts of Adsorption & Partition Chromatography; TLC, Paper, GC, GLC, HPLC, Ion exchange.

15 Lectures

UNIT-III

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Electrophoresis: Principle, Electrophoretic mobility(EPM) estimation, factors affecting EPM, Instrument design & set-up, Methodology & Applications of Free & Zone (Paper, Cellulose acetate, Agarose& Starch gel, Pulse-field, PAGE, SDS-PAGE, Capillary) Applications isoelectric focusing, 2D electrophoresis

15 Lectures

UNIT-IV

General biophysical methods-Measurement of pH, Radioactive labelling and counting, Autoradiography, X ray crystallography- X Ray diffraction, Bragg equation, Reciprocal lattice, Miller indices and unit cell, concept of different crystal structure, determination of crystal structure (Concept of rotating crystal method, powder method)

15

Lectures

BTH-52P-256 INSTRUMENTATION AND BIOTECHNIQUES - PRACTICAL

1. pH Meter- Standardization of pH meter.
2. Preparation of buffers.
3. Verification of Beer Lambert law using UV-Visible spectrophotometer.
4. Principle of TLC and Paper chromatography.
5. Column chromatography for protein /pigment.
6. Microscopy- compound Light microscope: principle, parts and functions.
7. Sterilization: principles and operations autoclave, Hot air oven, filtration, laminar air flow.
8. Principles and operations of Incubators & shakers.
9. Principles and operations of centrifuge.

Suggested readings:

1. **Biochemistry**-J. L. Jain
2. **Instrumental methods of chemical Analysis**-Chaitwal and Anand
3. **Biochemistry and Molecular Biology**-Wilson and Walker
5. **Biophysical Chemistry, Part II: Techniques for the study of biological structure and function**- Cantor & Schimmel
4. **The tools of Biochemistry**- Terrance G.Cooper
5. **Bioinstrumentation** - Veerakumari
6. **Biological Instrumentation and methodology** – Dr P K Bajpai
7. **Tools and techniques of biotechnology** – Mousumi Debnath
8. **Instrumental method of analysis in biotechnology** – Dinesh Kumar Chatanta
9. **Introduction to Instrumentation in Life sciences**-Prakash Singh Bisen, Anjana Sharma

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand the principles and applications of buffers in maintaining pH stability and the operational principles of pH meters.
2. Comprehend the principles and applications of various microscopy techniques including dissecting, compound, phase contrast, fluorescence, and electron microscopy (SEM and TEM).

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3. Gain knowledge of spectroscopic techniques including UV-visible spectrophotometry, IR spectroscopy, and Raman spectroscopy, focusing on their basic principles, instrumentation, and qualitative/quantitative applications.
4. Learn about fluorescence spectroscopy, centrifugation principles, types, and applications, and the basic concepts and techniques of chromatography (TLC, GC, HPLC, etc.).
5. Understand the principles and methodologies of electrophoresis, including various types (PAGE, SDS-PAGE, etc.), isoelectric focusing, 2D electrophoresis, and their applications in biotechnology.
6. Explore general biophysical methods such as radioactive labelling, autoradiography, and X-ray crystallography, including the concepts of X-ray diffraction, Bragg's equation, crystal structures, and methods for structure determination.

Raj Jais
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